

**Term 1 2013**

Mathematics Class:

[illegible]

SECTION I

5 marks

Attempt Questions 1 – 5

Use the multiple choice answer sheet for Questions 1 – 5.

1 Given that $\log_a x = 2.8$ and $\log_a y = 4.3$, the value of $\log_a \left(\frac{y^2}{ax} \right)$ is given by the expression

(A) $2 \times 4.3 - 2.8 + 1$

(B) $\frac{2 \times 4.3}{2.8 \times 1}$

(C) $4.3^2 - 2.8 \times 1$

(D) $2 \times 4.3 - 2.8 - 1$

2 Let $k = 3^x$. Which expression is equal to $\log_3 k^2$?

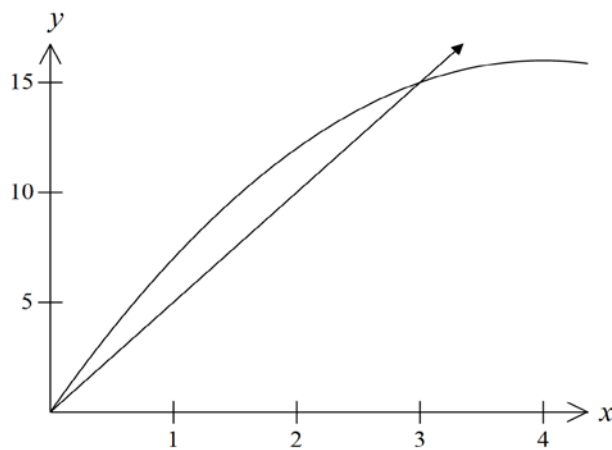
(A) 3^{2x}

(B) 3^{x^2}

(C) $2x$

(D) x^2

3 The diagram below shows the graph of $y = 5x$ and $y = 8x - x^2$.
The graphs intersect at (3, 15).



Which integral will give the area enclosed between $y = 5x$ and $y = 8x - x^2$?

(A) $\int_0^3 (3x - x^2) dx$

(B) $\int_0^3 (x^2 - 3x) dx$

(C) $\int_0^3 (13x - x^2) dx$

(D) $\int_0^3 (x^2 - 13x) dx$

- 4 The graph of $y = f(x)$ has been drawn to scale for $0 \leq x \leq 2$.

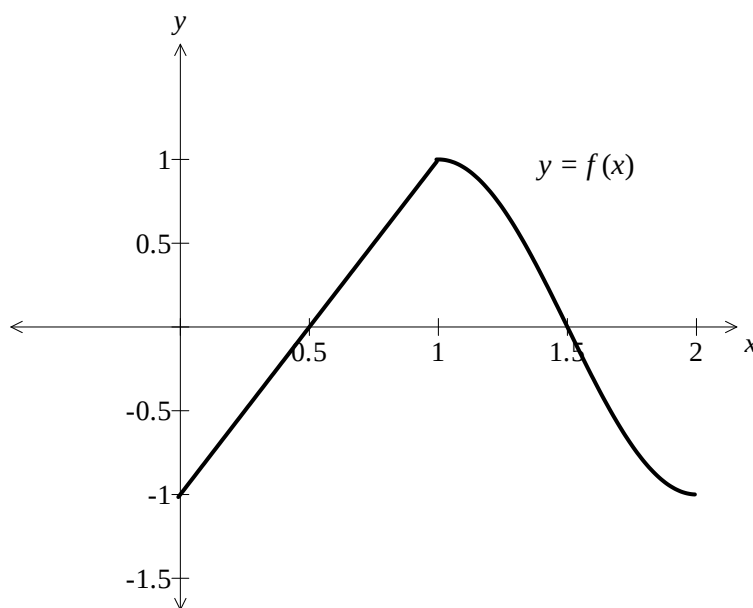
Which of the following has the greatest value?

(A) $\int_0^{0.5} f(x) dx$

(B) $\int_0^1 f(x) dx$

(C) $\int_0^{1.5} f(x) dx$

(D) $\int_0^2 f(x) dx$



- 5 The function $Q(n) = \frac{n}{6}(n+1)(4n+5)$ will sum the series

(A) $1 \times 3 + 2 \times 4 + \dots + n(n+2)$

(B) $1 \times 3 + 2 \times 5 + \dots + n(2n+1)$

(C) $1 \times 2 + 2 \times 3 + \dots + \frac{n}{2}(n+1)$

(D) $1 \times 2 + 2 \times 3 + \dots + n(n+1)$

SECTION II

40 marks

Attempt Questions 6 – 9

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 6 (11 marks) Use a SEPARATE writing booklet.

(a) Evaluate $\int_1^9 (1 + \sqrt{x}) \, dx$. 2

(b) Find $\int x \sqrt{1-x^2} \, dx$ by using the substitution $u = 1-x^2$. 3

(c) Use the substitution $u = 3-x$ to evaluate $\int_1^3 x(3-x)^4 \, dx$. 3

(d) Use mathematical induction to prove that 3

$$\frac{1}{1 \times 4} + \frac{1}{4 \times 7} + \dots + \frac{1}{(3n-2)(3n+1)} = \frac{n}{3n+1}$$

for all positive integers n .

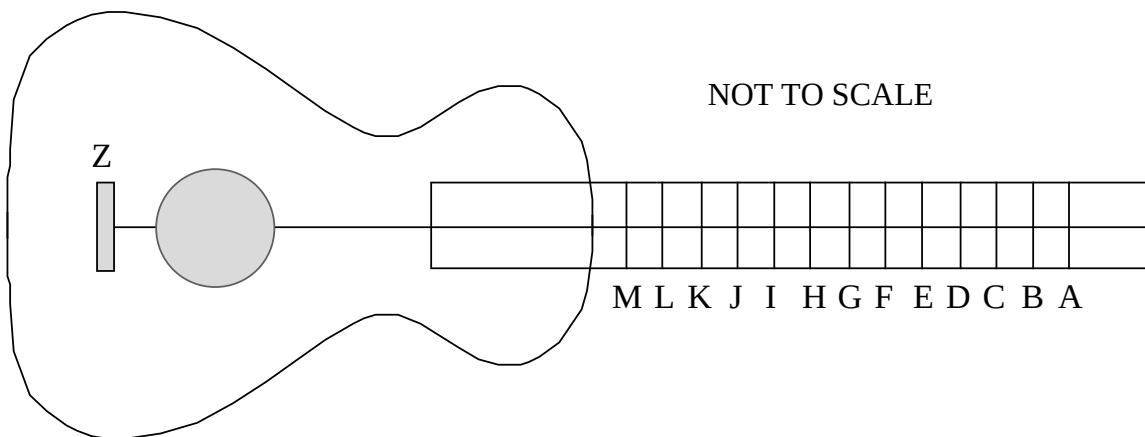
Question 7 (9 marks) Use a SEPARATE writing booklet.

(a) Find the least value of n such that $(0.95)^n < 0.5$, where n is an integer. 2

(b) (i) Expand $\sum_{n=1}^4 \left(1 - \frac{3^n - 2}{3^n}\right)$ and evaluate the sum. 2

(ii) Find $\sum_{n=1}^{\infty} \left(1 - \frac{3^n - 2}{3^n}\right)$. 1

(c) On the keyboard of a guitar, the mark M is the midpoint of A and Z.
The 13 marks lettered A to M are such that their distances from Z are terms in a geometric series.
The length AZ is 52 cm.



(i) Show that $r = \sqrt[12]{2}$. 2

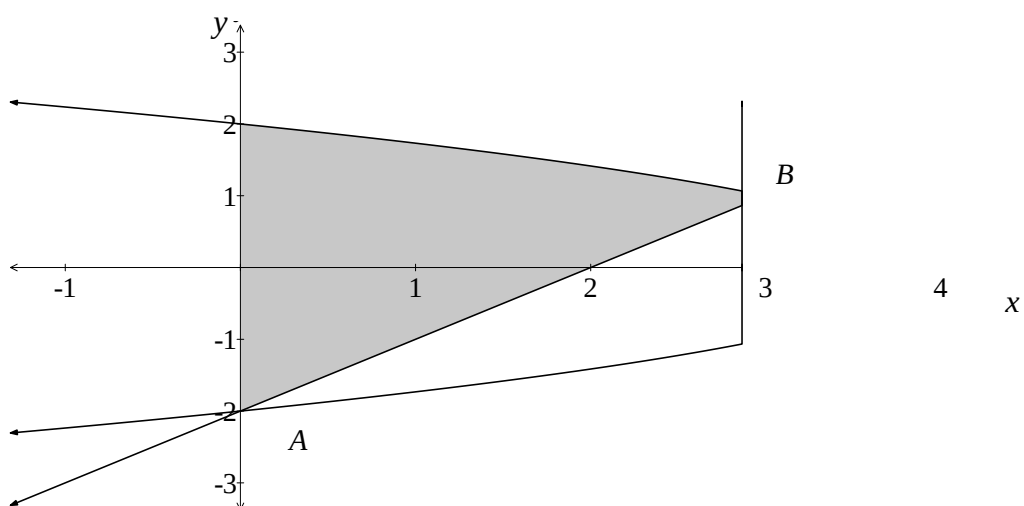
(ii) Find the distance FG correct to 1 decimal place. 2

Question 8 (8 marks) Use a SEPARATE writing booklet.

(a) (i) Differentiate $x\sqrt{x+3}$, leaving your answer as a simple fraction. 2

(ii) Hence find $\int \frac{x+2}{\sqrt{x+3}} dx$. 2

(b) The line $y = x - 2$ meets the sideways parabola $y^2 = 4 - x$ at two points $A(0, -2)$ and $B(3, 1)$ as shown.

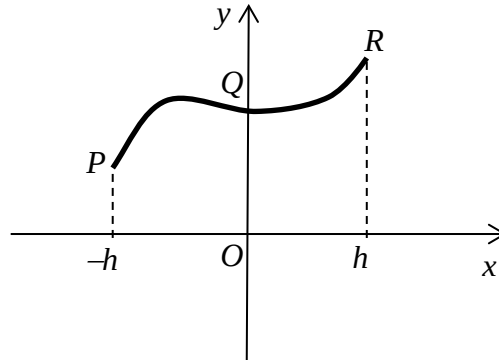


The shaded region is rotated about the y -axis.
Calculate the volume of the solid of revolution.

4

Question 9 (9 marks) Use a SEPARATE writing booklet.

- (a) Consider three points on the curve $y = ax^3 + bx^2 + cx + d$ with coordinates, $P(-h, y_0)$, $Q(0, y_1)$ and $R(h, y_2)$.



- (i) Find expressions for y_0 , y_1 and y_2 in terms of h, a, b, c and d . 2
- (ii) Hence write down an expression for $y_0 + 4y_1 + y_2$. 1
- (iii) Find a simplified expression for $\int_{-h}^h (ax^3 + bx^2 + cx + d) dx$, using calculus. 2
- (iv) Hence deduce that Simpsons Rule gives an exact value for the area bound by a cubic and the x -axis. 1
- (b) Use mathematical induction to prove that $5^n > 3^n + 4^n$ for all positive integers $n \geq 3$. 3

End of paper

Start here:

Solutions Ext 1 Task 2 2013

Multiple Choice

$$1. \log_{\frac{y^2}{ax}} = 2 \log_y y - (\log_a a + \log_a x)$$

$$= 2 \times 4 \cdot 3 - 1 - 2 \cdot 8$$

(D)

$$2. k = 3^x$$

$$\therefore \log_3 k = x$$

$$\therefore \log_3 k^2 = 2 \log_3 k$$

$$= 2x$$

(C)

$$3. \text{Area} = \int_0^3 (y_2 - y_1) dx$$

$$= \int_0^3 (8x - x^2) - 5x dx$$

$$= \int_0^3 3x - x^2 dx$$

(A)

4. $\int_0^{1.5} f(x) dx$ contains the greatest proportion above the x -axis. The value will be higher than any of the other integrals

(C)

$$5. Q(n) = \frac{n}{6} (n+1)(4n+5) \quad \text{where } Q(n) \text{ is the sum}$$

$$\text{Let } n=1 \quad Q(1) = 3 \quad \Rightarrow \text{either (A) or (B) could be right.}$$

$$n=2 \quad Q(2) = \frac{1}{3} \times 3 \times 13$$

$$= 13$$

$$\therefore (B) \text{ is correct since } 1 \times 3 + 2 \times 5 = 13$$

Start here :

Question 6

$$a) \int_1^9 (1 + \sqrt{x}) dx = \int_1^9 1 + x^{1/2} dx$$

$$= \left[x + \frac{2}{3} x^{3/2} \right]_1^9$$

$$= \left(9 + 18 \right) - \left(1 + \frac{2}{3} \right)$$

$$= 25\frac{1}{3}$$

$$b) u = 1 - x^2$$

$$\frac{du}{dx} = -2x$$

$$-\frac{1}{2} du = x dx$$

$$\int x \sqrt{1-x^2} dx = -\frac{1}{2} \int \sqrt{u} du$$

$$= -\frac{1}{2} \times \frac{2u^{3/2}}{3}$$

$$= -\frac{1}{3} u^{3/2} + C$$

$$= -\frac{1}{3} (1-x^2) \sqrt{1-x^2} + C$$

$$c) u = 3-x \Rightarrow x = 3-u$$

$$du = -dx$$

$$\text{When } x=3, u=0$$

$$\text{When } x=1, u=2$$

$$\therefore \int_1^3 x(3-x)^4 dx = -\int_2^0 (3-u) u^4 du$$

$$= \int_0^2 3u^4 - u^5 du$$

} reversing order
of limits

Question: _____

Student Number: _____

Start here :

$$= \left[\frac{3u^5}{5} - \frac{u^6}{6} \right]_0^2$$

$$= 8^{8/15} \text{ or } \frac{128}{15}$$

$$d) \frac{1}{1 \times 4} + \frac{1}{4 \times 7} + \dots + \frac{1}{(3n-2)(3n+1)} = \frac{n}{3n+1} \text{ for } n \in \mathbb{Z}^+$$

$$\text{Test result for } n=1 : \text{ LHS} = \frac{1}{1 \times 4} = \frac{1}{4}$$

$$\text{RHS} = \frac{1}{4}$$

$\therefore$ Result is true for $n=1$

Assume result holds for $n=k$

$$\text{i.e. assume that } \frac{1}{1 \times 4} + \frac{1}{4 \times 7} + \dots + \frac{1}{(3k-2)(3k+1)} = \frac{k}{3k+1}$$

We wish to show true for $n=k+1$

$$\text{i.e. show that } \frac{1}{1 \times 4} + \frac{1}{4 \times 7} + \dots + \frac{1}{(3k-2)(3k+1)} + \frac{1}{(3k+1)(3k+4)} = \frac{k+1}{3k+4}$$

$$\text{LHS} = \frac{k}{3k+1} + \frac{1}{(3k+1)(3k+4)} \text{ by assumption}$$

$$= \frac{k(3k+4) + 1}{(3k+1)(3k+4)}$$

$$= \frac{3k^2 + 4k + 1}{(3k+1)(3k+4)}$$

$$= \frac{(3k+1)(k+1)}{(3k+1)(3k+4)}$$

$$= \frac{k+1}{3k+4}$$

$\therefore$ By Mathematical Induction the result is proved true

Start here :

$$Q7 \ a) \ (0.95)^n < 0.5$$

$n = 14$ by trial and error on calc.

Alternatively, take logs of both sides :

$$n \log 0.95 < \log 0.5$$

$$n > \frac{\log 0.5}{\log 0.95} \quad \text{sign reverses because } \log 0.95 \text{ is negative}$$

$$n > 13.5$$

$\therefore n = 14$ is the least value of n .

$$b) \ (i) \ \sum_{n=1}^4 1 - \frac{3^n - 2}{3^n} = \sum_{n=1}^4 \frac{3^n - 3^n + 2}{3^n} \quad \left. \begin{array}{l} \text{Simplify} \\ \text{for ease of} \\ \text{working} \end{array} \right\}$$

$$= \sum_{n=1}^4 \frac{2}{3^n}$$

$$= \frac{2}{3} + \frac{2}{9} + \frac{2}{27} + \frac{2}{81}$$

$$= \frac{79}{81}$$

$$(ii) \ a = \frac{2}{3} \text{ and } r = \frac{1}{3}$$

$$S = \frac{a}{1-r}$$

$$= 1$$

Start here :

c) All distances are measured from Z

ie $AZ = 52 \text{ cm}$ and $MZ = 26 \text{ cm}$ Let $a = 26$ (MZ) in geometric seriesthen $\frac{LZ}{MZ} = r$ where r is the ratio

$$LZ = 26r$$

$\therefore AZ = 26r^{12}$ since there are twelve letters
after M (L, K, J, I, H, G, F, E,
D, C, B, A)

$$52 = 26r^{12}$$

$$r^{12} = 2$$

$$r = {}^{12}\sqrt{2}$$

$$(ii) FG = FZ - GZ$$

$$= 26r^7 - 26r^6$$

$$= 26(2^{7/12} - 2^{6/12})$$

$$= 2.186$$

$$= 2.2 \text{ cm}$$

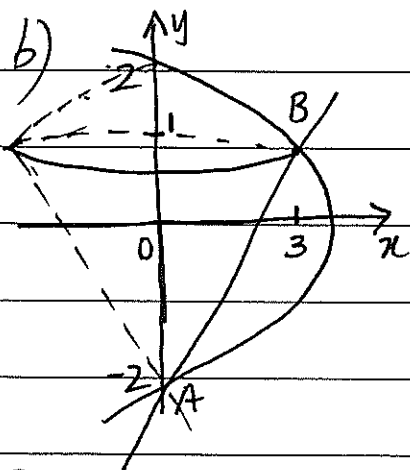
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Question 8

$$\begin{aligned}
 \text{a) } \frac{d}{dx} [x\sqrt{x+3}] &= \sqrt{x+3} + x \times \frac{1}{2} (x+3)^{-1/2} \times 1 \quad \text{product rule} \\
 &= \sqrt{x+3} + \frac{x}{2\sqrt{x+3}} \\
 &= \frac{2(x+3) + x}{2\sqrt{x+3}} \quad \left. \begin{array}{l} \text{simplify to} \\ \text{make work easier} \end{array} \right\}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{3x+6}{2\sqrt{x+3}} \\
 &= \frac{3(x+2)}{2\sqrt{x+3}} \quad \left. \begin{array}{l} \text{not in marking} \\ \text{scheme for (i)} \\ \text{Helpful for (ii)} \end{array} \right\}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii) } \int \frac{x+2}{\sqrt{x+3}} dx &= \frac{2}{3} \int \frac{3x+6}{2\sqrt{x+3}} dx \\
 &= \frac{2}{3} x\sqrt{x+3} + C
 \end{aligned}$$



Volume is considered in two parts:

 V_1 rotating the line about y axis betw $y=-2, 1$ V_2 rotating the curve about y axis betw $y=1, 2$

$$V = \pi \int_{-2}^1 (y^2 + 4y + 4) dy + \pi \int_1^2 (16 - 8y^2 + y^4) dy$$

Vol of cone

$$= \frac{\pi}{3} r^2 h$$

$$= \frac{\pi}{3} \times 3^2 \times 3$$

$$= 9\pi$$

$$= \pi \left[\frac{y^3}{3} + 2y^2 + 4y \right]_{-2}^1 + \pi \left[16y - \frac{8y^3}{3} + \frac{y^5}{5} \right]_1^2$$

$$= 9\pi + 3^{8/15} \pi$$

$$= 12^{8/15} \pi \text{ u}^3 \quad \text{or} \quad \frac{188\pi}{15} \text{ units}^3$$

Question 9

$$(i) \quad y_0 = a(-h)^3 + b(-h)^2 + c(-h) + d \\ = -ah^3 + bh^2 - ch + d$$

$$y_1 = d$$

$$y_2 = ah^3 + bh^2 + ch + d$$

$$(ii) \quad y_0 + 4y_1 + y_2 = 2bh^2 + 4d + 2d \quad \text{since odd powers} \\ = 2bh^2 + 6d \quad \text{add to give zero}$$

$$(iii) \quad \int_{-h}^h (ax^3 + bx^2 + cx + d) dx$$

$$= \left[\frac{ax^4}{4} + \frac{bx^3}{3} + \frac{cx^2}{2} + dx \right]_{-h}^h$$

$$= \left(\frac{ah^4}{4} + \frac{bh^3}{3} + \frac{ch^2}{2} + dh \right) - \left(\frac{ah^4}{4} - \frac{bh^3}{3} + \frac{ch^2}{2} - dh \right)$$

$$= \frac{2bh^3}{3} + 2dh$$

$$(iv) \quad \frac{h}{3} (y_0 + 4y_1 + y_2) \doteq \text{area under a curve by} \\ \text{Simpsons rule}$$

$$\frac{h}{3} (y_0 + 4y_1 + y_2) = \frac{h}{3} (2bh^2 + 6d) \quad \text{from (ii)} \\ = \frac{2bh^3}{3} + 2hd$$

$$= \int_{-h}^h (ax^3 + bx^2 + cx + d) dx$$

$\therefore$ Simpsons rule gives an exact value for the area bound by a cubic and x-axis

You may ask for an extra Insert if you need more space to answer questions.

Q9 b) Show $5^n > 3^n + 4^n$ for $n \geq 3$, $n \in \mathbb{Z}^+$

Test if true for $n=1$ LHS = $5^3 = 125$

$$\text{RHS} = 3^3 + 4^3$$

$$= 91$$

$\therefore$ Result is true for $n=3$

Assume result is true for $n=k$ i.e. assume $5^k > 3^k + 4^k$

We wish to show that $5^{k+1} > 3^{k+1} + 4^{k+1}$ is true

$$\text{LHS} = 5 \times 5^k$$

$$> 5(3^k + 4^k) \text{ by assumption}$$

$$= 5 \times 3^k + 5 \times 4^k$$

$$> 3 \times 3^k + 4 \times 4^k$$

$$= 3^{k+1} + 4^{k+1}$$

$$\therefore 5^{k+1} > 3^{k+1} + 4^{k+1}$$

$\therefore$ By mathematical induction result is true for all $n \geq 3$, $n \in \mathbb{Z}^+$

Alternatively consider $5^{k+1} - (3^{k+1} + 4^{k+1})$

$$> 5(3^k + 4^k) - 3 \cdot 3^k - 4 \cdot 4^k$$

$$= 2 \times 3^k + 2 \times 4^k$$

$$> 0$$

$$\therefore 5^{k+1} > 3^{k+1} + 4^{k+1}$$